

POST- EARTHQUAKE RETROFITTING IN NEPAL

22nd November 2016

Retrofitting Focus

- Cost efficiency potential of retrofitting earthquake damaged houses in 11 rural districts
- Further potential;
 - Kathmandu valley plus 17 additional rural districts
 - Homes already rebuild post-earthquake, not earthquake resistant
 - Homes in damage grades 1 & 2
 - Preventative retrofitting throughout Nepal

Retrofitting Potential

11 districts; eligible
beneficiaries only
100% of Grade 3
30% of Grade 4

Approx. 191,657 homes

From preliminary CBS damage
survey

Value for Money with Retrofit

Value for money for the homeowner;

1. DUDBC Design SMM 1.1 = 1,410 NPR/ft²
2. Average Retrofit = 292 NPR/ft²

$$1,410 \text{ NPR/ft}^2 = 15,197 \text{ NPR/m}^2; 292 \text{ NPR/ft}^2 = 3,145 \text{ NPR/m}^2$$

Living area 47 Sq.m.



Living area 108 Sq.m.

Comparison: New Build vs. Retrofit

Factor	Unit	New Build DUDBC design SMM 1.1*	New Build SMM 1.1* - RC Bands	Retrofit Lightweig ht Materials	Retrofit Reinforce d Concrete	Average Estimated Savings per Retrofit
Total Cost	NPR	771,254	719,095	360,112	319,210	405,514
Material	Tonne	145	165	18	28	132
Material	NPR	393,554	338,262	231,192	174,701	162,962
Demolition	NPR	50,000	50,000	0	0	50,000
Transport	NPR	23,200	26,333	2,920	4,509	21,052
Labour	Man Days	435	435	180	200	245
Labour	NPR	304,500	304,500	126,000	140,000	171,500
Duration	Weeks	14.5	14.5	6	6.7	8.2

* Discounting 50% salvage of stone and 20% salvage of wood

Cost Efficiency Potential with Retrofit

Approximate Savings for 191,657 HHs;

- ❑ 11.49 million metric tonnes less stone required
- ❑ 1.53 million man weeks less labour required
- ❑ 77.14 billion NPR (≈ 0.7 billion USD) reduction in reconstruction costs

Retrofitting Potential Included

11 districts; eligible beneficiaries
only

100% of Grade 3

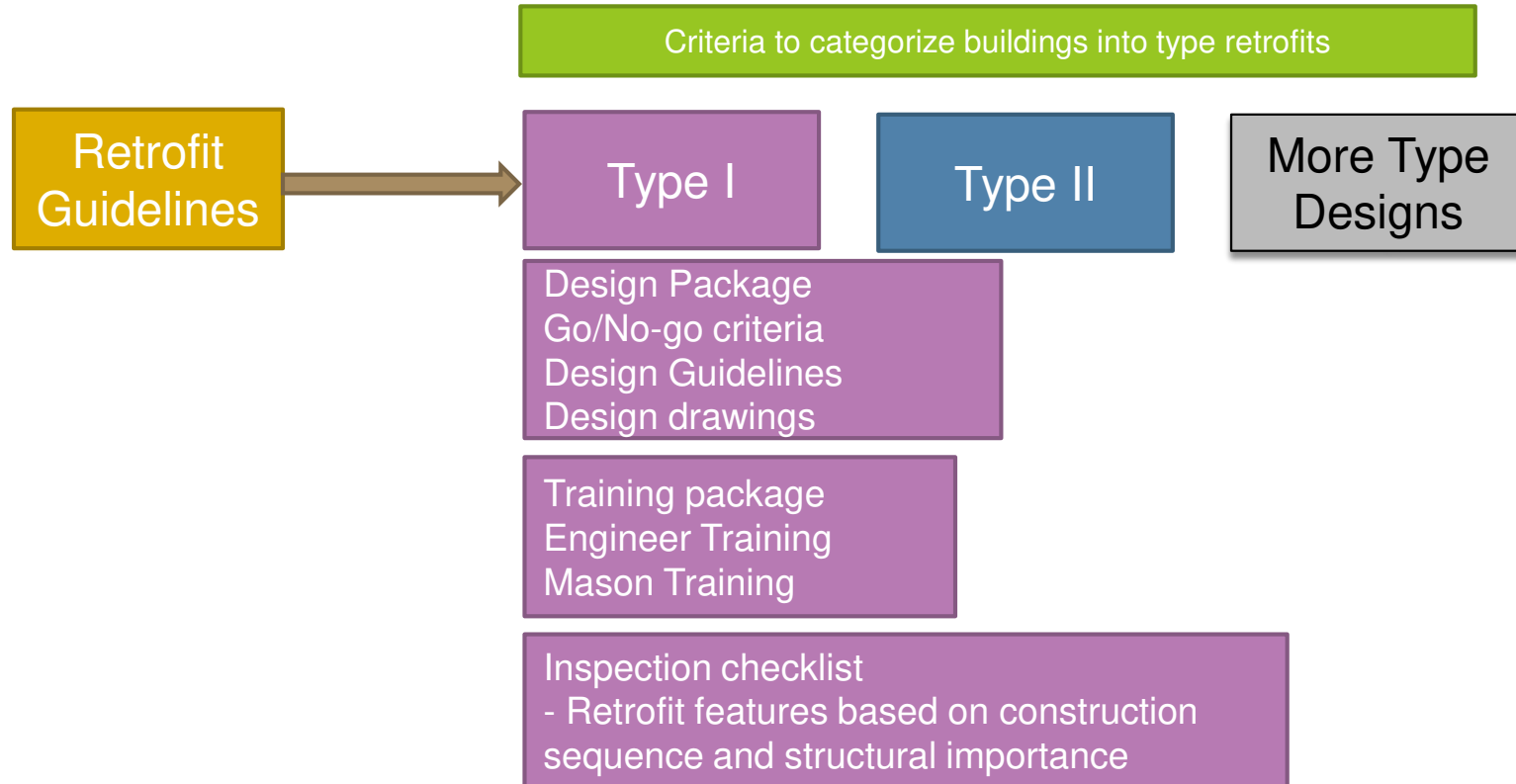
30% of Grade 4

Approx. 191,657 homes

Retrofit Type Design



Type Design



Type Design I: Go/No-go criteria

- Typical mud and stone buildings
- Building up to two storey and attic
- Timber floor (flexible floor) with mud finish
- Aspect ratio not greater than 3
- Longitudinal span not longer than 9 metres
- Transverse direction wall not longer than 4.5 metres
- Storey height not greater than 2.1 metre
- Attic wall height not greater than 1 metre
- Openings aligned at all floors
- Openings size not greater than 35% of the length of the wall
- Minimum length of pier to be 0.75 metre
- Minimum wall thickness to be 0.45 metre
- The walls of the building is not retaining soil on any side
- Building is not subject to other hazards
- Not applicable to buildings with significant distortion, especially to long walls and diaphragm

General Design Philosophy

- Use local materials and methods whenever possible.
 - Introduce as few changes as possible to achieve goal.
 - New materials should be available and cost effective.
- Mobilize strength of existing structure.
 - “Tie the building together”
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Typical Deficiencies

Typical Deficiencies in Stone in Mud Mortar Buildings



Gable End Wall Damage



Short wall Out-of-plane
Damage



Typical Deficiencies in Stone in Mud Mortar Buildings



Floor joists not tied at the ends, rests directly on wall



Delamination of walls

Positive Building Attributes

- Relatively thick walls
- Low storey heights
- Continuous main beam
- Independent post support for main beam at transverse walls
- Good joist embedment in longitudinal walls

Retrofit Process

Installation of Ring Beam at top of the wall

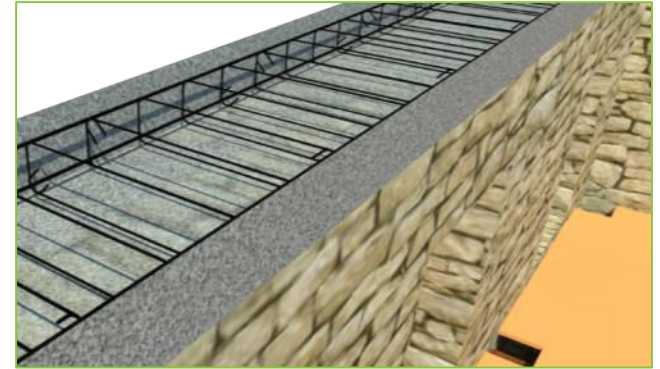


Figure:
Reinforcement
of Ring Beam



Roof Detail

Positive Connection of Roof to Ring Beam

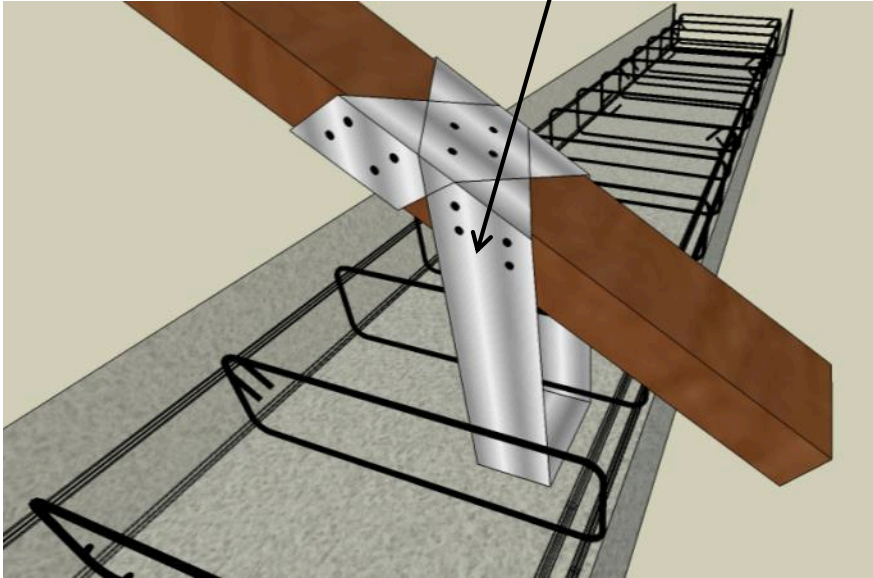


Rafter Connection
Before



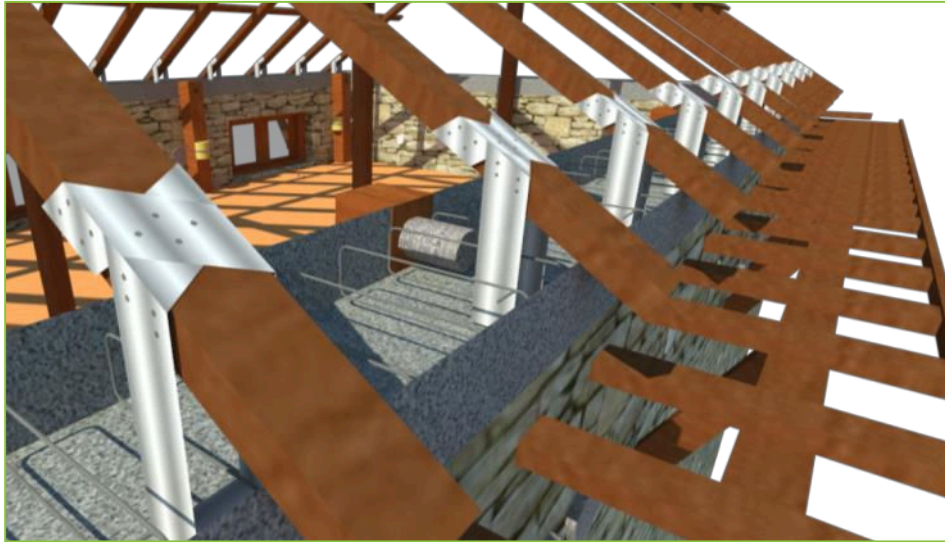
Positive Connection of Roof to Ring Beam

Strap embedded
in concrete to
bind rafter



Roof Detail

Positive Connection of Roof to Ring Beam



Roof Detail

Sill plate and bracing of rafters



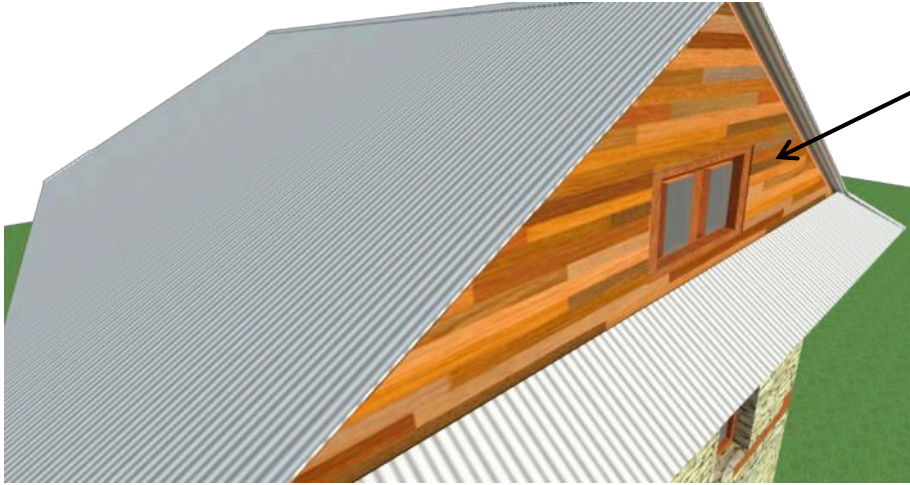
Binding Rafter and
Purlins with GI Wire

Roof Detail

Sill plate and bracing of rafters



Installation of Light Gable



*Stone Gable replaced
by light weight
wooden plank*



Installation of Through Stone



Figure showing Installation of Through Stone

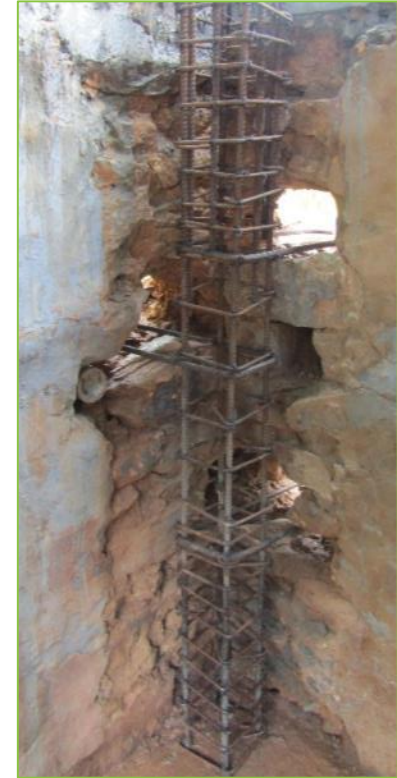
Through Stone

Installation of Strong Backs



*U-Hook connecting
Wall and Strong
Backs*

*Figure showing Strong Backs at
Corners*

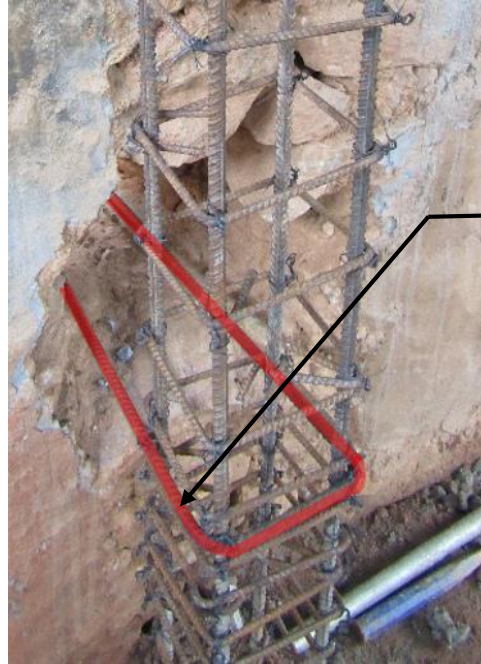


Strong Backs

Installation of Strong Backs



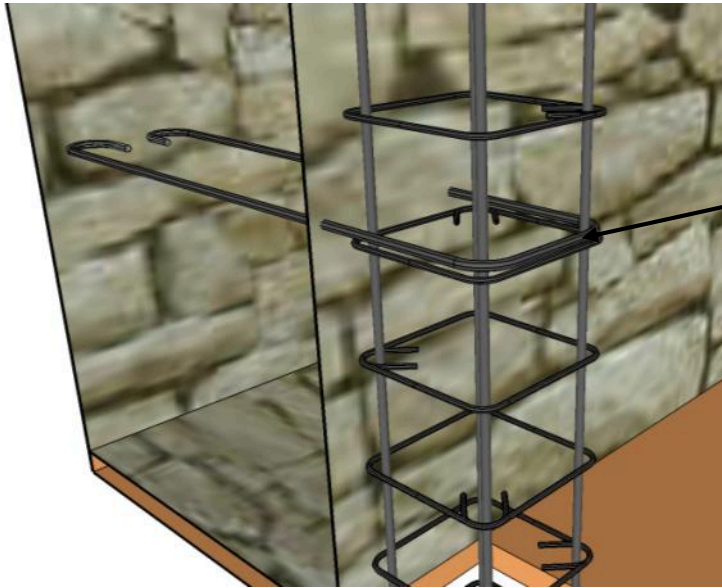
*Figure showing
Strong Backs at
Longitudinal wall*



*U-Hook
connecting Wall
and Strong
Backs*



Installation of Strong Backs

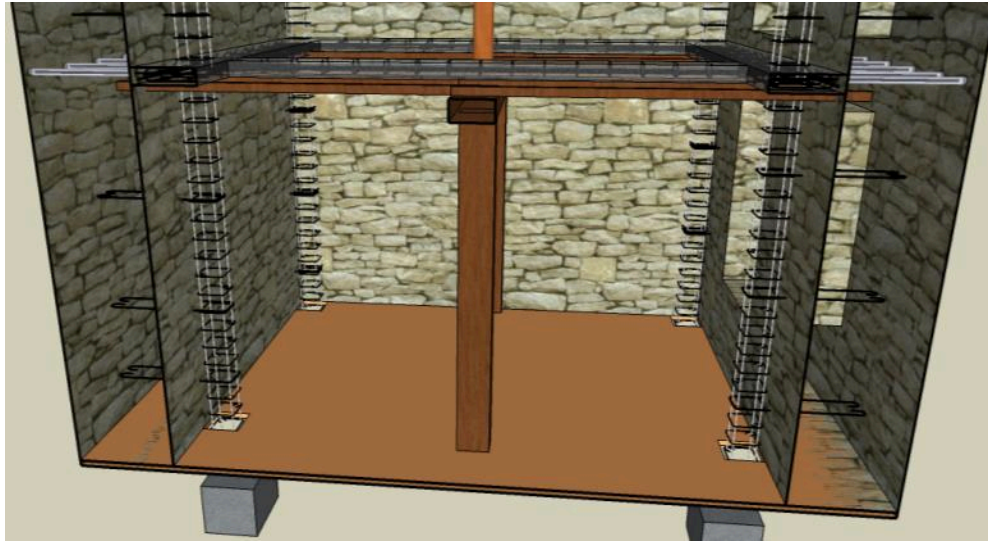


*U-Hook
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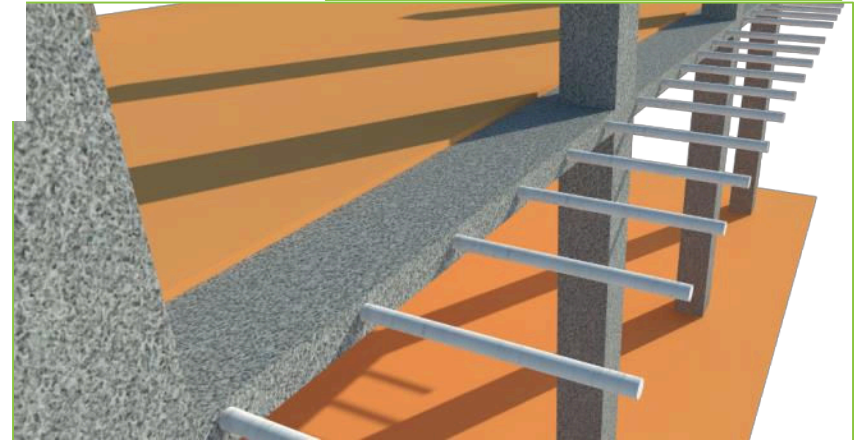
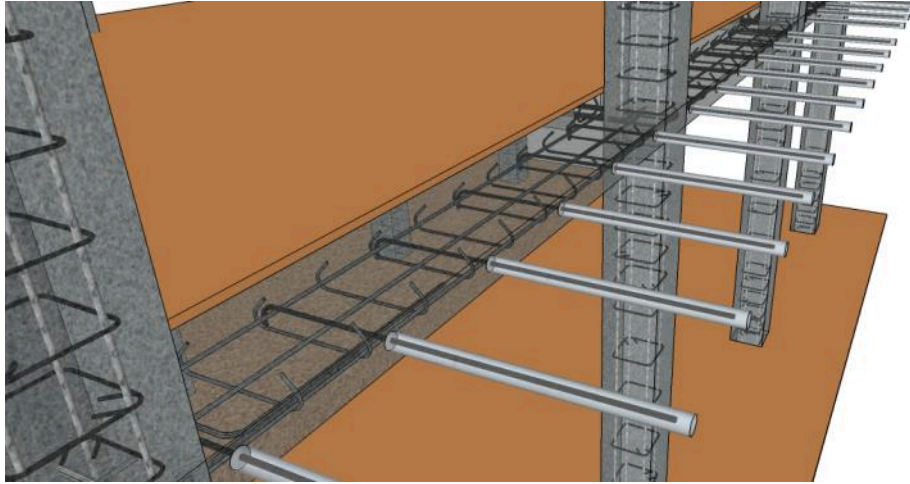
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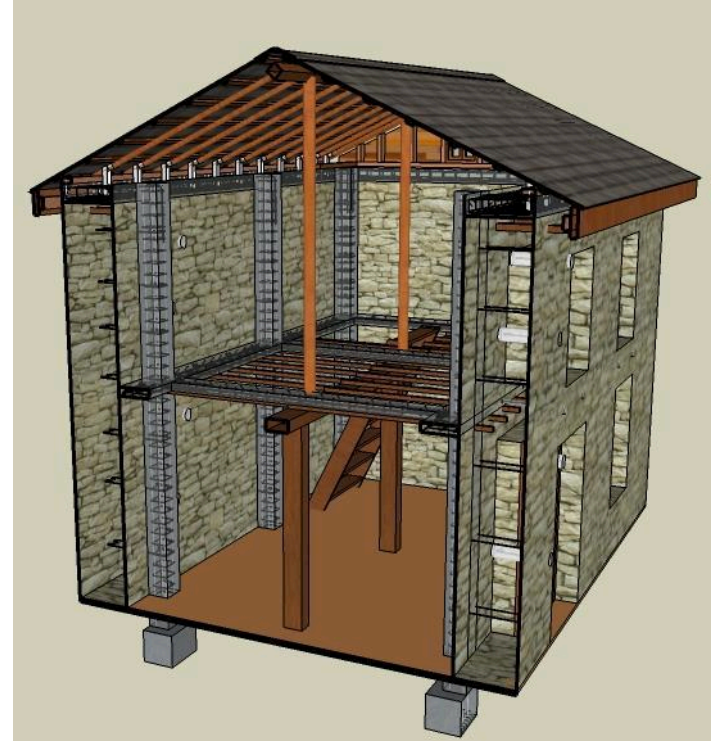
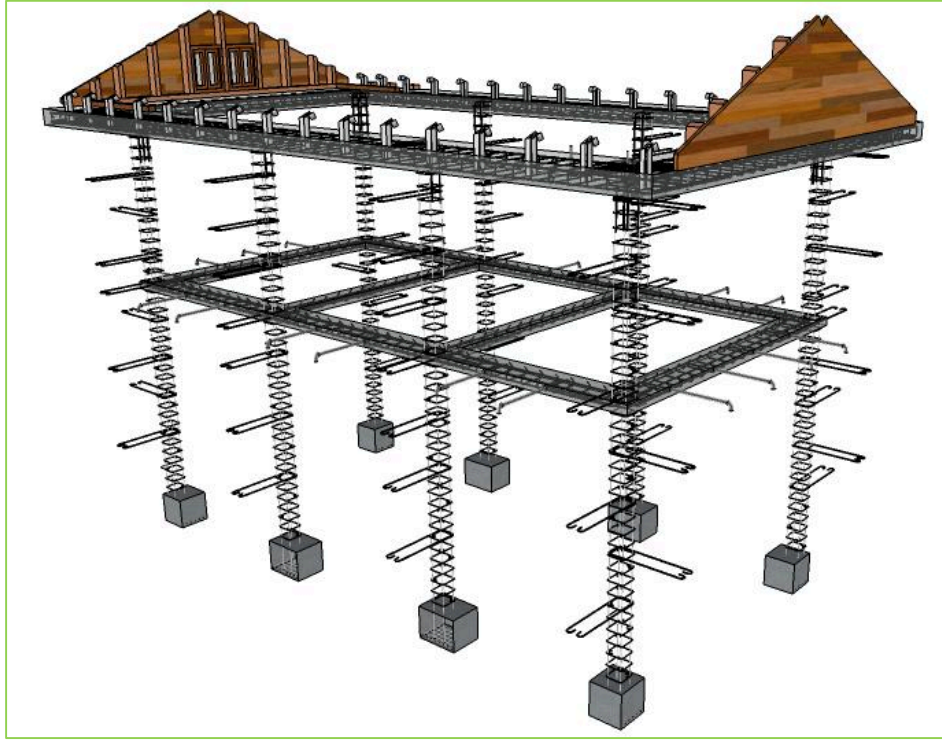
Connection of Strong Backs to the Wall



Strengthening of the Diaphragm



Retrofit elements



Plaster on the Inside and Outside of the Building



Retrofit (Timber)



Retrofit (Timber)



What Next

- The approval process for retrofit type designs
- How to scale this up to meet the huge potential of retrofits
 - Awareness on retrofitting for Engineers, masons, homeowners
 - Government endorsement of retrofitting
 - More Type designs
 - How to make these type designs accessible to engineers and homeowners
- Anything else?